



The utility of Random Forests for wildfire severity mapping

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ABSTRACT

Reliable fire severity mapping is a vital resource for fire scientists and land management agencies globally. Satellite derived pre- and post-fire differenced severity indices (Δ FSI), such as the differenced Normalised Burn Ratio (Δ NBR), are widely used to map the severity of large wildfires. Fire severity classification is commonly undertaken through the identification of severity class thresholds in Δ FSI. However, several shortcomings have been identified with severity classifications using Δ FSI, including poor delineation of low fire severity classes, and unsatisfactory performance when Δ FSI classification thresholds are applied to new landscapes. Our study assesses the performance of the Random Forest classifier (RF) for improving the accuracy of satellite based wildfire severity mapping across heterogeneous landscapes using Landsat imagery. We collected point based fire severity training data ($n = 10,855$) from sixteen large wildfires occurring across south-eastern Australia between 2006 and 2016. The predictive accuracy of fire severity classification using Δ NBR and the RF incorporating numerous spectral indices, was assessed using bootstrapping and cross validation. Image acquisition and index calculation for each fire was undertaken in Google Earth Engine (GEE). Results of the bootstrapping validation show that the RF classifier had very high classification accuracy ($> 95\%$) for unburnt (UB), crown scorch (CS) and crown consumption (CC) severity classes, and high classification accuracy ($> 74\%$) for low severity classes (crown unburnt, CU; partial crown scorch, PCS). The RF classification outperformed the Δ NBR classification for all severity classes, increasing classification accuracy by between 6%–21%. Cross validation using independent fires produced similar median classification accuracy as the bootstrapping validation, though the RF classification of CU was substantially reduced to $\sim 55\%$. Δ NBR, Δ NDWI and Δ NDVI were the three most important variables in the RF model. The Landsat satellite platform used to derive the indices had little effect on classification accuracy. Maps produced using the RF classifier in GEE had similar spatial patterns in fire severity classes as maps produced using time-consuming hand digitisation of aerial images. GEE was found to be a highly efficient platform for image acquisition, processing and production of severity maps. Our study shows that fire severity mapping using RF classifiers provides a robust method for broad scale mapping of fire severity across heterogeneous landscapes. Furthermore, the GEE-based classification framework supports the operational application of this approach in a land management agency context for the rapid production of fire severity maps.

1. Introduction

Fire is one of the dominant disturbances across terrestrial ecosystems globally (Bowman et al., 2009). Fire severity is defined as the loss of above- and below-ground organic matter and is correlated with fire intensity within plant communities with similar vegetation structure (Hammill and Bradstock, 2006; Keeley, 2009). Fire severity is an important component of the fire regime, as it influences the post-fire

response of plant and animal communities (Bennett et al., 2016; Smucker et al., 2005), alters erosion and water quality (Doerr et al., 2006; Nolan et al., 2015) and influences the likelihood of fire suppression by firefighting crews (Bradstock et al., 2010). Accurate fire severity mapping provides crucial information for policy makers, land managers and researchers (Eidenshink et al., 2007), as well as staff involved in incident control (e.g. post-fire impact assessments). In particular, fire severity mapping can facilitate: (i) burnt area emergency

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response (Keeley, 2009), (ii) assessment of decadal trends in fire regimes (Eidenshink et al., 2007), (iii) assessments of the effectiveness of fire management strategies (Price and Bradstock, 2012; Thompson et al., 2007), (iv) improved understanding of fire behaviour (Collins et al., 2014; Holden et al., 2009), and (v) insight into fire regime effects on biota and ecosystem processes (Bennett et al., 2016; Knox and Clarke, 2012; Smith et al., 2016). Broad scale fire severity mapping predominantly uses data from satellite imagery and aerial photography (Keeley, 2009; Kolden et al., 2015; McCarthy et al., 2017), although systematic and reliable mapping of severity is not a common process in many countries (Kolden et al., 2015).

Fire severity mapping using satellite imagery has largely relied on the pre- and post-fire differencing of a spectral index (SI) to derive a single differenced fire severity index (Δ FSI) (Eidenshink et al., 2007; Hammill and Bradstock, 2006; Veraverbeke et al., 2010). The rationale behind this approach is that pre- and post-fire differences in the SI will reflect the degree of environmental change due to fire, and hence fire severity. Δ FSI may be used as a continuous measure of fire severity (e.g. Collins et al., 2007), though often Δ FSI are classified into environmentally meaningful fire severity classes (e.g. Hammill and Bradstock, 2006). The preferred SI used to derive the Δ FSI will vary depending on the ecosystem characteristics or features of interest (e.g. foliage, soil). Δ FSI derived from SI sensitive to foliage health and cover, such as the differenced Normalised Burn Ratio (Δ NBR) (Boer et al., 2008), tend to show a good correlation with field derived measures of fire severity (Hammill and Bradstock, 2006; Parker et al., 2015; Veraverbeke et al., 2010). Consequently, Δ NBR is a widely utilised index for fire severity mapping (Lentile et al., 2006), and is favoured by many fire management agencies (e.g. Kolden et al., 2015). Despite its widespread use, Δ NBR has been shown to be sub-optimal for describing fire severity in some instances (Roy et al., 2006).

The approach of fire severity classification using a single Δ FSI has a number of limitations that restrict the application of the method across space and time. First, Δ FSI values are not comparable across structurally distinct vegetation classes (e.g. forest and shrubland; Hammill and Bradstock, 2006; Miller et al., 2009; Parker et al., 2015). This is because vegetation structure influences SI values (Boer et al., 2008), hence the maximum value of a Δ FSI will be determined by pre-fire vegetation state (Brewer et al., 2005; Miller et al., 2009; Roy et al., 2006). Consequently, accurate severity classification with a single Δ FSI will require adjustments to the thresholds for each vegetation type examined (Brewer et al., 2005; Hammill and Bradstock, 2006), which reduces the broad scale usefulness of the approach. Second, low fire severity classes, or classes that are a composite of fire severity effects (e.g. partial crown scorch), can be difficult to distinguish with a single Δ FSI (Hammill and Bradstock, 2006; Miller et al., 2009). This partially reflects limitations in the type of change that a single SI can detect and difficulties in distinguishing pixels with mixed cover types (Miller et al., 2009). Fire alters a range of ecosystem properties, including foliage cover, health and moisture content and the amount of bare soil and charcoal; each which influence spectral reflectance (White et al., 1996). Classification methods using empirical models that can incorporate information from multiple Δ FSI sensitive to a range of environmental conditions as well as SIs related to pre-fire vegetation conditions may facilitate improved fire severity classification (e.g. Brewer et al., 2005). Physical based models, such as Radiative Transfer Models that simulate spectral signatures at the leaf and canopy scale, have also been successfully used to predict burn severity (e.g. Composite Burn Index) from satellite imagery (e.g. De Santis et al., 2009), and may outperform Δ FSI such as Δ NBR (De Santis et al., 2010).

Machine learning (ML), defined as the application of statistical techniques and algorithms for identifying patterns in data and making predictions from those patterns, are now commonly applied in remote sensing classification (Graves et al., 2016; Heydari and Mountrakis, 2018; Rogan et al., 2008). ML techniques perform better than simple classifiers in dealing with complex interactions between scene

complexity, scale and aggregation and have improved discrimination of classes in heterogeneous landscapes (typical in remote sensing) with low inter-class separability and high intra-class variability (Ghimire et al., 2012). Random Forest (RF) (Breiman, 2001) is a popular machine learning technique, commonly applied in remote sensing (Belgiu and Drăguț, 2016). Frequently cited advantages of RF classifiers over other machine learning techniques include its excellent classification results and processing speed (Belgiu and Drăguț, 2016; Du et al., 2015), its ability to handle noise and outliers in complex measurement space (Mellor et al., 2015; Rodriguez-Galiano et al., 2012), and characterize complex variable interactions (Cutler et al., 2007). The RF classifier is well suited to addressing fire mapping problems (e.g. Meddens et al., 2016; Ramo and Chuvieco, 2017), as it can consider multiple environmental variables simultaneously (Hultquist et al., 2014; Meddens et al., 2016), and has been found to be better suited for fire severity mapping than other ML classifiers (Hultquist et al., 2014). Despite the commonly cited advantages of ML classifiers like RF, they have rarely been utilised for fire severity classification (Barrett et al., 2011; Hultquist et al., 2014; Meddens et al., 2016).

Studies assessing the utility of ML classification techniques for fire severity mapping have been limited by the size and extent of training and validation data, and have not performed cross validation using fires that were not included in training data (e.g. Brewer et al., 2005; Hultquist et al., 2014; Meddens et al., 2016). Consequently, the true utility of ML classifiers, such as RF, as a tool to reliably produce fire severity mapping across heterogeneous landscapes has not been assessed. The aim of our study was to:

- (i) assess the performance of RF for classifying fire severity using several indices derived from moderate resolution (30 m) Landsat imagery;
- (ii) compare the RF classification to a single index classification using Δ NBR, both within and outside the set of training fires;
- (iii) assess the relative importance of different indices in improving fire severity classification; and
- (iv) examine how additional sampling effort from new fires can improve RF classification accuracy of these fires.

2. Methods

2.1. Study area

The study included sixteen large wildfires, ranging from 1800 to > 120,000 ha in extent, that occurred between 2006 and 2016 in Victoria, Australia (Fig. 1). The wildfires used in this study were selected based on the availability of (i) high resolution (≤ 35 cm) visible and infrared aerial photography captured shortly after the fire ignition date (i.e. within ~2–3 months) and (ii) Landsat imagery to produce pre- and post-fire cloud free mosaics. Fire severity was spatially heterogeneous within the wildfire perimeters, covering the range of fire severity classes distinguishable with aerial photography (Table 1, SM1). The fires occurred in a range of vegetation communities including heathlands (i.e. shrub dominated communities), open forests and woodlands, tall forests and pine plantations. Grass and sedge dominated communities were excluded from the study, as were arid and semi-arid woody vegetation communities.

2.2. Fire severity data

Fire severity data used in this study for classifier training and validation was derived using aerial photo interpretation and digitisation. High resolution visible and infrared aerial photography has been effectively used to distinguish fire severity classes (Table 1) in Australian forests, woodlands and shrublands, with high correlations being observed between photo and field based measures of fire severity (Hammill and Bradstock, 2006; McCarthy et al., 2017). Five fire

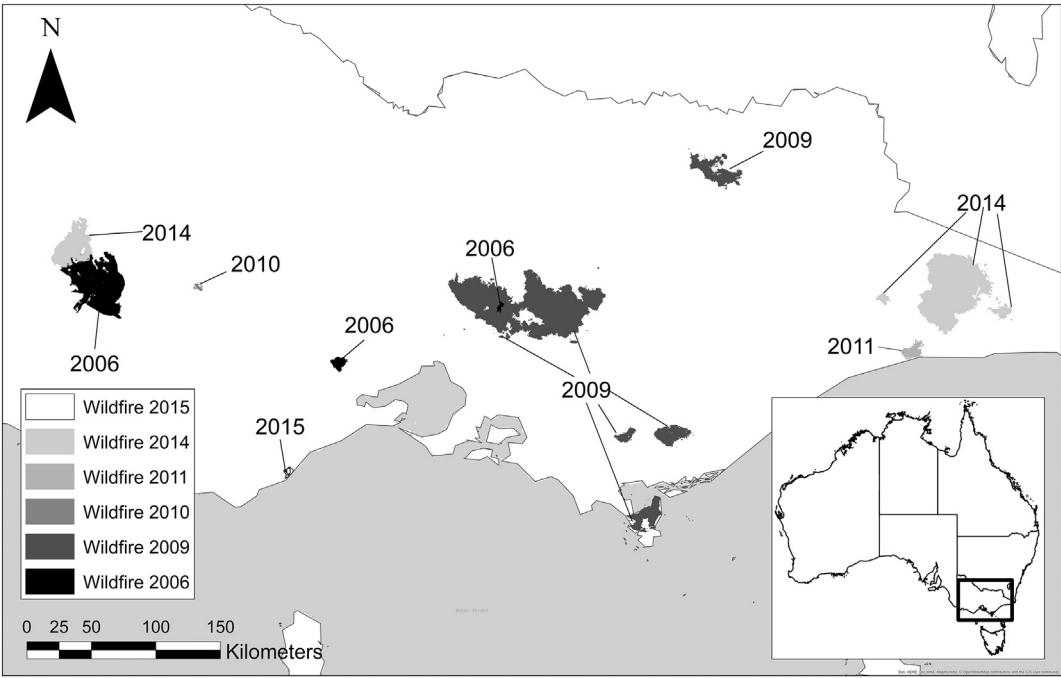


Fig. 1. The location of the sixteen wildfires examined across Victoria, Australia.

severity classes were identified from aerial photographs, based on levels of scorch and consumption of the crowns of the canopy and understorey vegetation layers (Table 1; Fig. S1). These classes reflect different degrees of organic matter loss (i.e. foliage and litter consumption) (Hammill and Bradstock, 2006) and are indicative of the potential for fire suppression in eucalypt forests and woodlands (Bradstock et al., 2010). These five fire severity classes are currently used for routine mapping using aerial photography conducted by Victoria's Department of Environment, Land, Water and Planning (DELWP) (McCarthy et al., 2017). Similar fire severity classification schemes based on organic matter loss have been used elsewhere in Australia (e.g. Chafer et al., 2004; Hammill and Bradstock, 2006) and Internationally (see Keeley, 2009).

Post-fire infrared aerial imagery was obtained from DELWP for each of the sixteen study fires. Large patches (> 100 m wide) of each fire severity class were identified across each wildfire and digitised in ArcMap (v10.2.2) by one observer (L. Collins), to ensure there was no inter-observer bias. Unburnt patches were generally identified outside the fire perimeter, typically within 2000 m of the fire edge, as large patches of unburnt vegetation within the fire perimeter were not always present. Digitised polygons covered a broad range of environmental

conditions (e.g. vegetation type, logging history, fire history) within each wildfire perimeter to ensure a range of conditions were included. The scale of the Landsat pixel (30 m) was considered when assigning areas to fire severity classes, by placing size and spacing restrictions on fire severity patterns. In areas assigned to the partial crown scorch (PCS) class, which is a mixture of unburnt crown (CU) and crown scorch (CS) (Table 1), the distance between patches of CU and CS could not exceed ~25 m to ensure a composite of fire severity classes would occur within a Landsat pixel. In areas assigned as unburnt (UB), CU and CS, patches of non-target severity classes could not be greater than ~5 m radius and were typically > 25 m apart to ensure they didn't occupy > 10% of a Landsat pixel. Distances were measured in ArcMap using the Measure tool.

Random sample points were generated for model training within each digitised severity polygon, such that points were > 50 m from the polygon edge, to ensure we were sampling Landsat pixels that fell entirely within the target severity class. Sample points were located > 85 m apart to ensure separation of at least one Landsat pixel and to minimise the effect of spatial autocorrelation. A total of 10,855 points were generated across the sixteen fires (Table 1), which included samples of each fire severity class across the majority of fires (n = 15;

Table 1
Fire severity classes identified from aerial photographs in this study. Descriptions of the visual appearance of aerial photos and the damage to vegetation are provided. Visual examples of the fire severity classes are provided in Fig. S1.

Severity class	Appearance in aerial photos	Vegetation damage*	Sample points (n)
Crown consumption (CC)	> 10% canopy foliage consumed. Appears black in the infrared photographs, with small patches of scorched crown foliage.	Complete consumption of foliage in the lower and taller shrub layer. Near complete consumption of foliage in the tree canopy layer.	2836
Crown scorch (CS)	> 90% of canopy foliage is scorched. Appears brown in infrared photographs	Lower and mid shrubs burnt. Near complete scorch of taller shrubs and the tree canopy	2878
Partial crown scorch (PCS)	Canopies have a combination of both unburnt and scorched foliage (10–90%). Appears red and brown in infrared photographs	Lower shrubs burnt, some scorching of taller shrubs and the tree canopy.	947
Crown unburnt (CU)	> 90% of canopy foliage is unburnt, but the understorey has been burnt. Canopy is red and understorey is black or brown in infrared photographs	Surface fire with variable litter consumption. Taller shrubs and tree canopy remain unburnt.	1428
Unburnt (UB)	> 90% of canopy and understorey foliage is unburnt. Canopy and understorey are red in infrared photographs	Litter and vegetation largely unaffected by fire	2766

Table 2

Spectral indices used for fire severity classification in this study. Indices considered for the RF classification included pre- and post-fire differenced indices (denoted by Δ) and pre- and post-fire indices (denoted by 'pre' or 'post'). References in the Environmental characteristics column are examples from Australian ecosystems.

Index	Equation	Reference	Environmental characteristics	Indices considered
NBR	$\frac{NIR - SWIR2}{NIR + SWIR2}$	Key and Benson (2006)	Sensitive to foliage health and leaf area index (Boer et al., 2008)	NBR_{pre} , NBR_{post} , ΔNBR
NDVI	$\frac{NIR - R}{NIR + R}$	Tucker (1979)	Sensitive to foliage health and leaf area index (Coops et al., 1997)	$NDVI_{pre}$, $NDVI_{post}$, $\Delta NDVI$
NDWI	$\frac{NIR - SWIR1}{NIR + SWIR1}$	Gao (1996)	Sensitive to live foliage moisture content (Caccamo et al., 2012)	$NDWI_{pre}$, $NDWI_{post}$, $\Delta NDWI$
VARI	$\frac{G - R}{G + R - B}$	Gitelson et al. (2002)	Sensitive to live foliage moisture content (Caccamo et al., 2012; Nolan et al., 2016)	$VARI_{pre}$, $VARI_{post}$, $\Delta VARI$
BAI	$\frac{1}{(0.1 - R)^2 + (0.06 - NIR)^2}$	Chuvieco et al. (2002)	Emphasises charcoal signal post fire	BAI

Table S1).

2.3. Remotely sensed indices

Five commonly used SI derived from Landsat imagery (5 TM, 7 ETM +, 8 OLI/TIRS) were investigated as potential predictors of fire severity in this study, including the Normalised Burn Ratio (NBR), Normalised Difference Vegetation Index (NDVI), Normalised Difference Water Index (NDWI), Visible Atmospherically Resistant Index (VARI) and the Burn Area Index (BAI) (Table 2). These SI were selected because they are commonly used for fire detection and severity mapping (Hawbaker et al., 2017; Veraverbeke et al., 2010), due to their sensitivity to a range of ecosystem characteristics that are altered by fire, including foliage health and cover (NBR, NDVI), foliage moisture content (NDWI, VARI) and charcoal (BAI) (Table 2). Pre- and post-fire SI differencing was used for NBR, NDVI, NDWI and VARI. BAI was calculated using only post-fire images. Pre- and post-fire SI values were also calculated for NBR, NDVI, NDWI and VARI (Table 2).

Google Earth Engine (GEE) was used for image acquisition and processing, and the derivation of spectral indices. GEE is a cloud-based remote sensing data processing and analysis platform, which combines public access remote sensing and geospatial data, with high-performance parallel computation service (Gorelick et al., 2017). Landsat surface reflectance products, which are acquired from the United States Geological Survey (USGS), were used in our study. All scenes have been georeferenced and terrain corrected (Level T1) to a level deemed satisfactory for research purposes. The surface reflectance product uses the Landsat Ecosystem Disturbance Adaptive Processing System (LEDAPS) algorithm to correct for atmospheric effects on reflectance (Schmidt et al., 2013). The surface reflectance product also contains a cloud, cloud shadow and water mask (CFmask), which can be used to mask out water bodies or areas affected by clouds (<https://landsat.usgs.gov/>).

Fire start and finish dates were derived from digital archives (i.e. fire history mapping, online reports), or estimated using Landsat or MODIS imagery. Post-fire imagery acquired within the ~70 days following the burn finish date was considered suitable for the post-fire SI, as previous work has shown that the fire severity signal in Australian woodlands and heathlands begins to deteriorate beyond this window (Parker et al., 2015). Imagery acquired within 60 days prior to the fire start date was considered suitable to calculate the pre-fire SI, to minimise seasonal differences on the pre- and post-fire SI. If single cloud-free pre- and post-fire imagery were available within the sampling window defined above, they were used to calculate indices, with preference given to the cloud free images captured closest to the fire start and end dates. Composites of images were used when suitable cloud free images were not available. Composites were derived by calculating SI for each image falling within the desired date range, with cloud and shadow affected pixels being masked out of each image using the CFmask. The shortest possible window either side of the fire start and end dates was used to obtain a cloud free composite. Water was masked out of pre-fire images using the CFmask, but not post-fire images, as the

CFmask often confused severely burnt patches as water. A median index value was then calculated from all suitable pixels to derive the composite. Fires occurring prior to February 2013 used both Landsat 5 TM and Landsat 7 ETM+ imagery, while those occurring after this time used only Landsat 8 OLI/TIRS imagery. Landsat 8 imagery was not composited with Landsat 7 imagery due to subtle differences in band widths (Roy et al., 2016).

2.4. Modelling and analysis

2.4.1. Random forest

RF uses bootstrap aggregated (bagging) sampling of training data to construct decision trees (base classifiers), which represent a set of diverse base classifiers (Mellor and Boukir, 2017), which when combined into an ensemble are used to assign a class (prediction) to a response variable through voting (in the case of categorical data) or by averaging (for continuous variables) (Breiman, 2001). In constructing each decision tree in the RF, at every node (split point), a randomly selected subset of model predictor variables (parameter *mtry*) is evaluated for partitioning training data into increasingly homogenous subsets, the predictor variable used to split the training data is that which results in the highest increase in data uniformity. The parameter *ntree* represents the number of decision trees (base classifiers) in the RF ensemble.

2.4.2. Comparison of classification approaches and assessment of index importance

We initially trialled the RF classification using different combinations of indices, including all indices, without post-fire FSI and without ΔFSI (S3). Estimates of error rates for the different models were all within 0.4% (Table S3), indicating little difference in model classification accuracy. For the purpose of parsimony, we focused on a RF model containing nine SI, including the ΔFSI (NBR, NDVI, NDWI, VARI) and their pre-fire index values and BAI. The satellite identifier (i.e. Landsat 8 or Landsat 5/Landsat 7) was included to determine if the narrower Landsat 8 band widths affected classification. The number of trees was set to 500 and number of predictor variables at each node used was the square root of the number of variables used in the model. Bootstrapping was used to obtain estimates of the median and variance in classification accuracy of the RF classifiers (Champagne et al., 2014). A subset of 70% of the data was randomly selected without replacement and used to train the RF classifier. The random selection was stratified by fire severity class to ensure that data used for training and validation has equal proportions of each severity class. The trained RF was used to classify the remaining 30% of the dataset in order to provide independent model validation. The process was repeated randomly 1000 times and the proportion of correctly classified data points in the validation dataset was recorded. Classification accuracy of the bootstrapped RF classifiers was graphically presented using violin plots, which depict both a boxplot and the probability density of the data.

Fire severity classification using single ΔFSI typically uses thresholds in the index to identify fire severity classes (Kolden et al., 2015). We trialled two approaches to perform this classification: (i) by

calculating mean Δ FSI values for each fire severity class, and using the mid-points between the mean values of the ordered severity classes as thresholds for classification [see Hammill and Bradstock, 2006]; and (ii) using ordinal regression (see Harris et al., 2011). Classification performance for each Δ FSI was assessed using the bootstrapping approach described above. Δ NBR was found to have the highest classification accuracy (~80%) of the Δ FSI using both approaches (Table S4). We used the conceptually simple midpoint threshold Δ NBR classification as the baseline to compare the RF classification against. The midpoint threshold Δ NBR classification had similar overall classification accuracy as the ordinal regression approach (78.9% and 80.8% respectively). Differences in the accuracy of the RF and Δ NBR classification were statistically assessed for each severity class by extracting the 95% confidence intervals (CI) from the bootstrapped data. Non-overlapping CI were determined to be significantly different ($P < 0.05$).

A leave one fire out cross validation was undertaken for the RF classifier and the Δ NBR method by excluding one fire from the training dataset and using all the sample points from that fire for validation. We repeated this process so that each fire was used only once for validation ($n = 16$). In each validation fire, classification accuracy was only assessed for fire severity classes that had > 20 observations, to ensure reliable accuracy estimates were obtained. The accuracy of the RF and Δ NBR classification was graphically displayed using violin plots. A Wilcoxon signed rank test was used to test for differences ($P < 0.05$) between the accuracy of the RF and Δ NBR classifiers within fire severity classes.

We assessed how the addition of sampling points from a validation fire influenced the accuracy of the RF classifier, by incrementally adding data from a validation fire (10, 20, 30, 40, and 50 points per fire severity class) to the training dataset and revalidating the predictive accuracy of the RF classifier. The validation process was only conducted using four wildfires with a minimum of 75 data points per severity class, to facilitate the removal of up to 50 points per severity class from the validation dataset. Samples included in the training data were taken randomly without replacement from the validation fire and this process was only repeated 200 times to reduce processing time.

The full dataset was used to train the RF classifier and the Gini impurity criterion was extracted to identify the relative influence of different indices on fire severity classification. We assessed the influence of the top five indices (I_1 – I_5) by fitting five models that were derived by incrementally adding variables based on the Gini importance (i.e. I_1 , $I_1 + I_2$, $I_1 + I_2 + I_3$, $I_1 + I_2 + I_3 + I_4$, $I_1 + I_2 + I_3 + I_4 + I_5$). The bootstrapping process described above, where 70% and 30% of the data was used for model training and validation, was used to estimate model accuracy.

Statistical analysis was undertaken in R v 3.4.1 (R Development Core Team, 2016). Random Forest analysis was undertaken using the *randomForest* package (Liaw and Wiener, 2002). Classification matrices were derived using the *caret* package (Kuhn, 2008). Violin plots were produced using the package *vioplot* (Adler, 2005).

2.4.3. Fire severity mapping

Classified fire severity maps were produced in GEE using the Random Forest classifier to provide visual examples of model performance. The number of trees was set to 100, as the modelling described above showed that 100 trees was sufficient to produce a stable error rate. The number of predictor variables supplied to each node equalled the square root of the number of variables in the model. Maps produced in GEE were visually compared to available severity maps derived from digitised aerial photographs (McCarthy et al., 2017).

3. Results

The RF model showed a high degree of classification accuracy (91%) for the bootstrapping cross validation (Fig. 2). Median classification accuracy was 95% or greater for the unburnt (UB), crown scorch

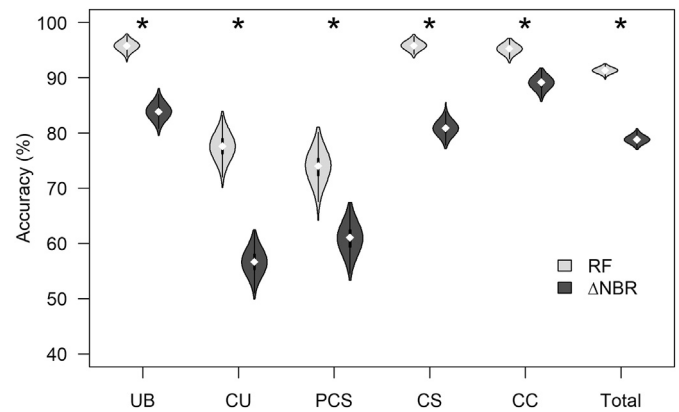


Fig. 2. Violin plots depicting the accuracy (%) within severity classes for the RF and Δ NBR classification when applied to validation data taken from within the fires used for classifier training. Training data consisted of a random sample (70%) of the full dataset and validation was conducted on the remaining data. The training and validation process was repeated 1000 times. The plots display both the probability density of the data (shaded area), median (white diamond), interquartile range (black rectangle), and the first and third quartile $\pm 1.5 \times$ the interquartile range (black vertical lines) of the 1000 iterations. Non-overlapping 95% confidence intervals are denoted with an asterisk. Fire severity class acronyms are defined as follows: UB – unburnt; CU – crown unburnt; PCS – partial crown scorch; CS – crown scorch; CC – crown consumption.

(CS) and crown consumption (CC) fire severity classes and ~75% for the crown unburnt (CU) and partial crown scorch (PCS) classes (Fig. 2). The RF model had significantly ($P < 0.05$) greater classification accuracy than the Δ NBR threshold classification for all fire severity classes (Fig. 2), and showed good improvement (10–20%) in classification accuracy relative to Δ NBR for the UB, CU, PCS and CS classes (Fig. 2, Table 3). Relative to the Δ NBR threshold classification, the RF approach

Table 3

Classification matrix showing the difference (%) in classification accuracy between the RF and Δ NBR threshold classification. Median values and lower (L) and upper (U) 95% confidence intervals (in parentheses) from the 1000 bootstrap iterations are presented. Values highlighted in bold font along the diagonal show change in the correct classification of the validation data, whereby positive values reflect greater accuracy using the RF model. Values depicted in normal font show change in the misclassification of the validation data, whereby negative values reflect lower misclassification using the RF model. Values presented in italics have 95% confidence intervals that do not overlap zero, indicating significant change. Empty cells had too few observations ($n < 250$ iterations) to calculate a meaningful median and CI. Fire severity class acronyms are defined as follows: UB – unburnt; CU – crown unburnt; PCS – partial crown scorch; CS – crown scorch; CC – crown consumption.

Predicted	Observed				
	RF - Δ NBR				
	UB	CU	PCS	CS	CC
UB	11.81 (L: 9.52; U: 14.33)	–10.72 (L: –14.22; U: –7.00)	–0.71 (L: –2.11; U: 0.36)	–	–
CU	–11.93 (L: –14.33; U: –9.52)	20.75 (L: 16.32; U: 25.18)	–10.88 (L: –15.08; U: –6.32)	–	–
PCS	0.00 (L: –0.12; U: 0.36)	–9.09 (L: –12.12; U: –6.28)	13.33 (L: 7.72; U: 17.90)	–13.43 (L: –15.51; U: –11.34)	0.00 (L: –0.23; U: 0.23)
CS	–	–	–1.75 (L: –4.91; U: 2.10)	15.04 (L: 12.73; U: 17.25)	–6.11 (L: –7.87; U: –4.70)
CC	–	–	0.00 (L: 0.00; U: 0.70)	–1.04 (L: –2.08; U: 0.00)	6.11 (L: 4.7; U: 7.87)

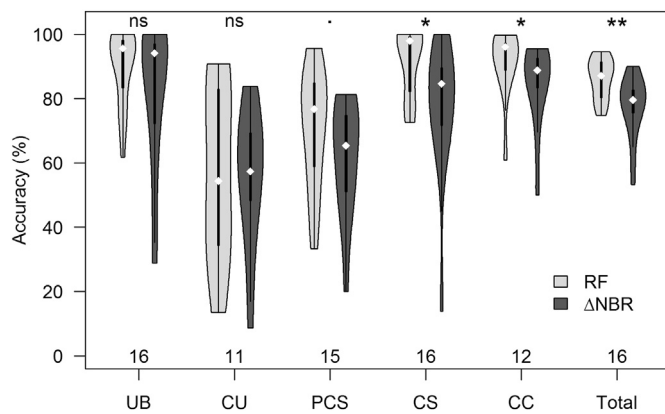


Fig. 3. Violin plots depicting the accuracy (%) within severity classes for the RF and Δ NBR classification when applied to new fires. The plots display both the probability density of the data (shaded area), median (white diamond), interquartile range (black rectangle), and the first and third quartile $\pm 1.5 \times$ the interquartile range (black vertical lines). Numbers along the x-axis are the number of fires that were used in the 'leave one out' cross validation. Results of the paired Wilcoxon signed rank test are shown above the violin plots with the following codes: $P > 0.1$ (ns), < 0.1 – 0.05 (.), < 0.05 – 0.001 (*) and < 0.001 (**). Fire severity class acronyms are defined as follows: UB – unburnt; CU – crown unburnt; PCS – partial crown scorch; CS – crown scorch; CC – crown consumption.

reduced misclassification between all adjacent severity classes (Table 3). Most notable were the reductions in misclassification between the UB and CU classes, CU and PCS classes, and the PCS and CS classes (Table 3).

The median predictive accuracy of the RF classifier for the leave one fire out cross validation (Fig. 3) was similar to values reported for the bootstrapping cross validation (Fig. 2), suggesting that the RF classifier will likely have high predictive accuracy when the model is applied to new fires. The exception was the CU class, which had a median accuracy of $\sim 55\%$ when applied to new fires (Fig. 3). However, the range of predictions for any fire severity class increased when RF was applied to new fires (i.e. the probability density plots increase in length; Fig. 2 vs Fig. 3), indicating that severity classes within some fires were poorly predicted. This was particularly evident for the CU and PCS fire severity classes (Fig. 3). Relative to the Δ NBR classification, the RF classifier had 5%–15% greater median predictive accuracy for the PCS, CS and CC classes (Fig. 3). Both approaches performed poorly for the CU class (Fig. 3). The addition of training data from the validation fire improved classification accuracy of the RF classifier in the CU class (Fig. 4), but had little effect on the other classes. The addition of 50 new training data points increased the median accuracy in the CU class by $\sim 20\%$ (Fig. 4), with median accuracy approaching that of the initial validation presented in Fig. 2.

The Gini statistic for the RF classifier identified Δ NBR as the variable with the greatest influence on classification accuracy, with Δ NDWI and Δ NDVI also having high importance (Fig. 5). The satellite platform used to derive the indices had little effect on classification accuracy (Fig. 5), suggesting that the small differences in bandwidth were unimportant. There was little difference in the predictive accuracy of the RF classifier using only Δ NBR versus the RF using the full suite of indices for CC (Fig. 6). Classification accuracy for the CU and PCS classes was poor when only using Δ NBR, and improved by $> 20\%$ with the inclusion of additional Δ FSI, with Δ NDWI and Δ NDVI being particularly influential (Fig. 6). The addition of Δ NDWI improved classification accuracy of UB and CS classes by $\sim 10\%$ (Fig. 6). RF classification accuracy using only Δ NBR (72%) was lower than classification accuracy with Δ NBR thresholds.

Example maps produced using Landsat imagery with the RF classifier in GEE showed comparable patterns in fire severity classification to

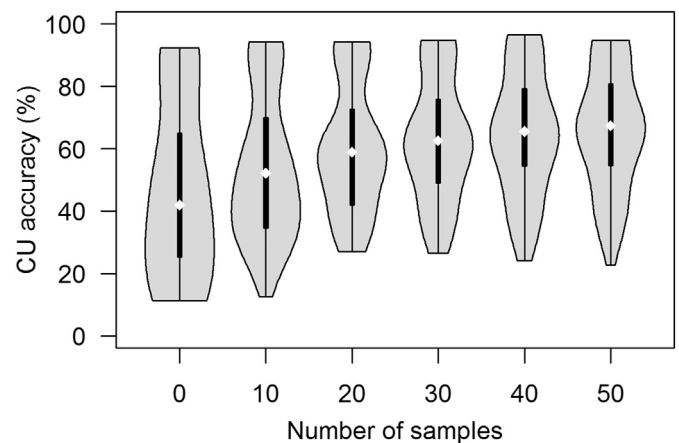


Fig. 4. The effect of adding new sample data to the RF classifier on the predictive accuracy of the crown unburnt (CU) severity class. The violin plots display both the probability density of the data (shaded area), median (white diamond), interquartile range (black rectangle), and the first and third quartile $\pm 1.5 \times$ the interquartile range (black vertical lines).

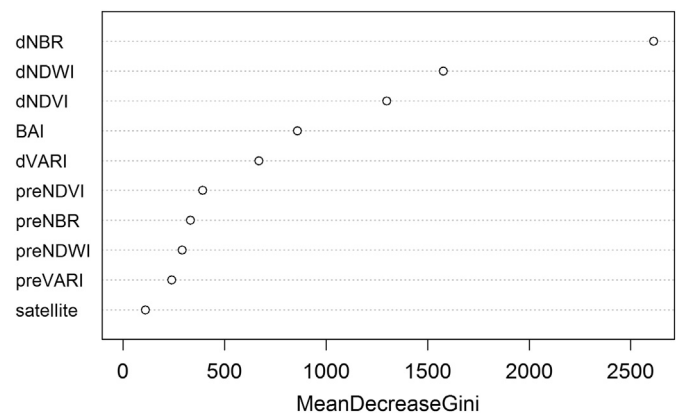


Fig. 5. Change in variable importance for the RF fitted to the full dataset, as measured by the mean decrease in Gini scores. Larger values denote greater importance.

hand digitisation of aerial photographs (McCarthy et al., 2017) (Fig. 7, S5).

4. Discussion

Our paper provides one of the first robust assessments of the utility of the RF algorithm for wildfire severity mapping relative to single Δ FSI methods that are widely used by fire management agencies globally (e.g. Kolden et al., 2015). Fire severity classification using the RF classifier showed significant improvement relative to classification using only Δ NBR, with a $\sim 10\%$ increase in overall classification accuracy when training data was provided for a fire. Significant improvements in classification accuracy were observed for all fire severity classes ($\sim 6\%$ – 21%), with considerably less misclassification between low to moderate fire severity classes (i.e. UB, CU, PCS, CS). Based on these findings, we recommend that the RF classifier, or similar classification techniques that can incorporate many spectral indices, be used for fire severity classification in favour of single Δ FSI approaches.

Previous assessments of ML techniques for fire severity classification have shown improved classification accuracy relative to single Δ FSI (Brewer et al., 2005; Hultquist et al., 2014; Meddens et al., 2016), suggesting that ML classifiers are likely to improve severity classification when applied to most ecosystems. Our results suggest that improved classification with the RF classifier largely arose from the ability

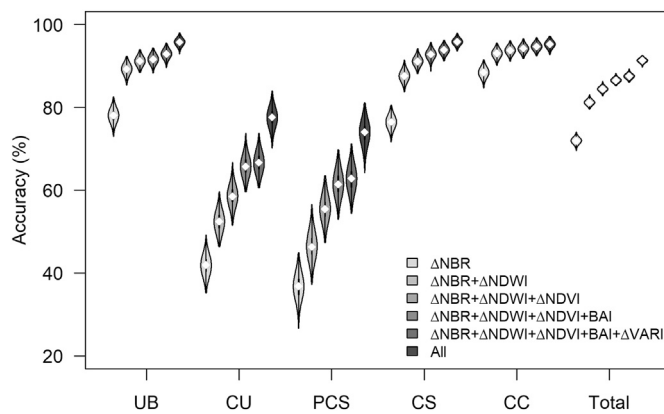


Fig. 6. Violin plots depicting the accuracy (%) of the RF classifier within severity classes in response to the addition of Δ FSI and SI. In each severity class violin plots are ordered from the RF classifier with the least (left) to most (right) number of variables. Training data consisted of a random sample (70%) of the full dataset and validation was conducted on the remaining data. The training and validation process was repeated 1000 times. The plots display both the probability density of the data (shaded area), median (white diamond), inter-quartile range (black rectangle), and the first and third quartile $\pm 1.5 \times$ the interquartile range (black vertical lines) of the 1000 iterations. Fire severity class acronyms are defined as follows: UB – unburnt; CU – crown unburnt; PCS – partial crown scorch; CS – crown scorch; CC – crown consumption.

to incorporate many Δ FSI sensitive to a range of environmental attributes (i.e. foliage cover, foliage moisture and charcoal) (Figs. 5, 6). The addition of pre-fire SI to provide context of vegetation states led to additional, albeit smaller, improvements. The value of incorporating pre-fire SI has been demonstrated previously for fire severity classification across structurally diverse ecosystems (Brewer et al., 2005).

The RF and Δ NBR approaches displayed high classification accuracy of crown fires, irrespective of whether the training data included information from the validation fire(s) (Fig. 2) or not (Fig. 3). This was expected as previous research has shown Δ NBR is very effective for identification of high severity fire (Miller et al., 2009). Δ NBR is highly sensitive to leaf area index, which shows large pre- and post-fire differences in areas subjected to crown consuming fires (Boer et al., 2008). In low to moderate fire severity classes, changes to live foliage cover are more subtle, and the effects of structural heterogeneity of vegetation will create noise in the spectral signal, potentially reducing classification accuracy when using a single Δ FSI (Miller et al., 2009). Our results show that incorporating additional Δ FSI substantially improves classification accuracy in the low - moderate fire severity classes. The Δ NDWI, which is sensitivity to foliage moisture content, was particularly important for the classification of low - moderate fire severity classes (UB, CU, PCS, CS; Fig. 6). Stem heating can lead to cavitation and deformation of stem xylem, reducing transfer of water to foliage (Ducrey et al., 1996), which may explain why a SI sensitive to foliage moisture content was useful in delineating low severity classes.

The need to improve the accuracy of mapping for ecologically important fire severity classes has been identified previously (Meddens et al., 2016). Unburnt refuges are of particular importance for the persistence and post-disturbance recovery of plant and animal communities, and consequently are of high conservation value (Robinson et al., 2013). The RF approach had high median classification accuracy ($> 95\%$) for unburnt vegetation, which is similar to results reported for fires in the north-western United States (Meddens et al., 2016). Our study and that of Meddens et al. (2016) show that the RF classifier approach to fire severity mapping substantially improves the classification accuracy ($\sim 10\%$) of unburnt vegetation relative to threshold classifications using only Δ NBR. Improved fire refuge mapping will facilitate greater understanding of fire refuge dynamics, and the processes driving the spatial and temporal patterns of this important

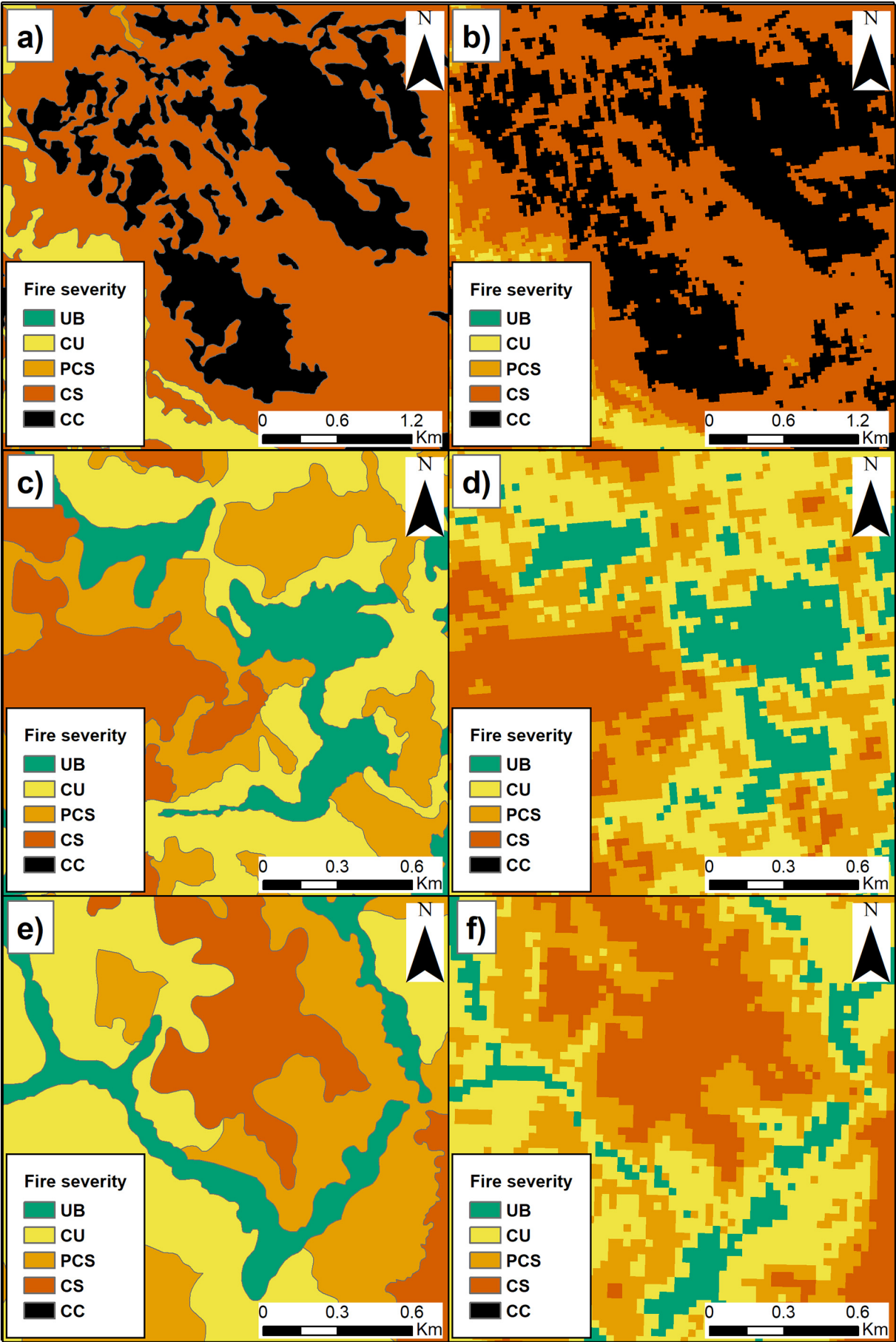
landscape feature (e.g. Krawchuk et al., 2016; Leonard et al., 2014).

The intra-fire validation approach used in our study highlights that the ability of the RF algorithm to classify the severity of entirely novel fire events may be overestimated if validation is only performed on data derived from within the training fires (Figs. 2; 3). This was particularly evident for the lower fire severity classes (CU, PCS), which present the greatest classification challenge. The issue of reduced predictive performance of RF when applied to independent datasets has been observed previously (e.g. Cutler et al., 2007). Past studies assessing the predictive accuracy of ML classifiers for fire severity classification (e.g. Hultquist et al., 2014; Meddens et al., 2016) have not performed cross validation on new fire events, and therefore may overestimate the predictive performance of the respective approaches. However, our work shows that the median classification accuracy of the RF algorithm on new fires still exceeded 95% for the UB, CS and CC fire severity classes, and was marginally better than the Δ NBR classification for the PCS class, indicating that the approach represents an improvement over traditional classification methods. Furthermore, we found that the addition of a small amount of training data (30–50 points) from the target fire could markedly improve the performance of the RF classifier for low fire severity classes (e.g. CU).

Our study found very high classification accuracy using a small set of spectral indices that are known to correlate well with environmental characteristics that are modified by fire (Table 2). The ‘optimal’ set of spectral indices for RF classification will vary depending on ecosystem characteristics. For example, our results, and the work of others (e.g. Meddens et al., 2016), suggest that incorporating indices that are sensitive to change in foliage cover and moisture content will be particularly useful for severity mapping in forest ecosystems with relatively continuous vegetation cover. However, in ecosystems with low vegetation cover, such as open woodlands and shrublands, fire severity may not correlate as well with indices sensitive to change in foliage (e.g. Δ NBR, Δ NDVI, Δ NDWI) (Epting et al., 2005; Veraverbeke et al., 2011). In the absence of pre-fire imagery, classification may be enhanced through the use of a broader suite of post-fire fire-specific indices, such as Normalised Burn Ratio Thermal, VI6T, Mid-Infrared Burn Index and Char Soil Index, which have been found to perform well for burnt area mapping and in some instances fire severity classification (Harris et al., 2011; Holden et al., 2005; Trigg and Flasse, 2001).

Although our study utilised classified fire severity data, the RF approach can also be applied to continuous measures of fire severity, such as the Composite Burn Index (CBI) (e.g. Hultquist et al., 2014). RF has produced better fits to CBI data than regression techniques utilising spectral indices (Hultquist et al., 2014). However, it is not clear whether RF will result in improved fire severity prediction relative to sophisticated simulation approaches, such as Radiative Transfer Models (Chuvieco et al., 2006; DeSantis and Chuvieco, 2007), which have been shown to have higher correlation with CBI than indices such as Δ NBR (De Santis et al., 2010). A key benefit of the RF approach presented in our study is that it only relies on training data and satellite derived spectral data, whereas simulation models such as RTM require additional parameterisation using field derived spectral properties of the environment (e.g. soil, charcoal, vegetation) (Chuvieco et al., 2006). The parameterisation of RTMs can be challenging (Yebara et al., 2018), and this may limit the uptake of these models by management agencies undertaking fire severity mapping. However, the benefit of RTMs is that they are physical models, thus offering greater generality than empirical models (e.g. Yebara et al., 2018) such as RF.

Spatial autocorrelation is an inherent feature of fire severity data (Collins et al., 2014; Taylor et al., 2014), and may lead to inflated predictive accuracy in RF models (Millard and Richardson, 2015). The nested sampling design used in our study, whereby sample points were generated inside polygons of homogeneous fire severity, artificially creates spatial autocorrelation, thus we could not meaningfully examine this in our study. Few studies have examined spatial autocorrelation of fire severity in Australian temperate forests, though the



(caption on next page)

Fig. 7. Examples of fire severity maps produced by the digitisation of aerial photography (left column) and Landsat images using the RF algorithm in GEE (right column). Rows depict a, b) high severity fire, c, d) low severity fire with coarse areas of unburnt vegetation and e, f) low severity fire with fine areas of unburnt vegetation. Boundaries of severity classes derived from the digitised aerial photos (left column) are smoother as they are a vector format. Fire severity class acronyms are defined as follows: UB – unburnt; CU – crown unburnt; PCS – partial crown scorch; CS – crown scorch; CC – crown consumption.

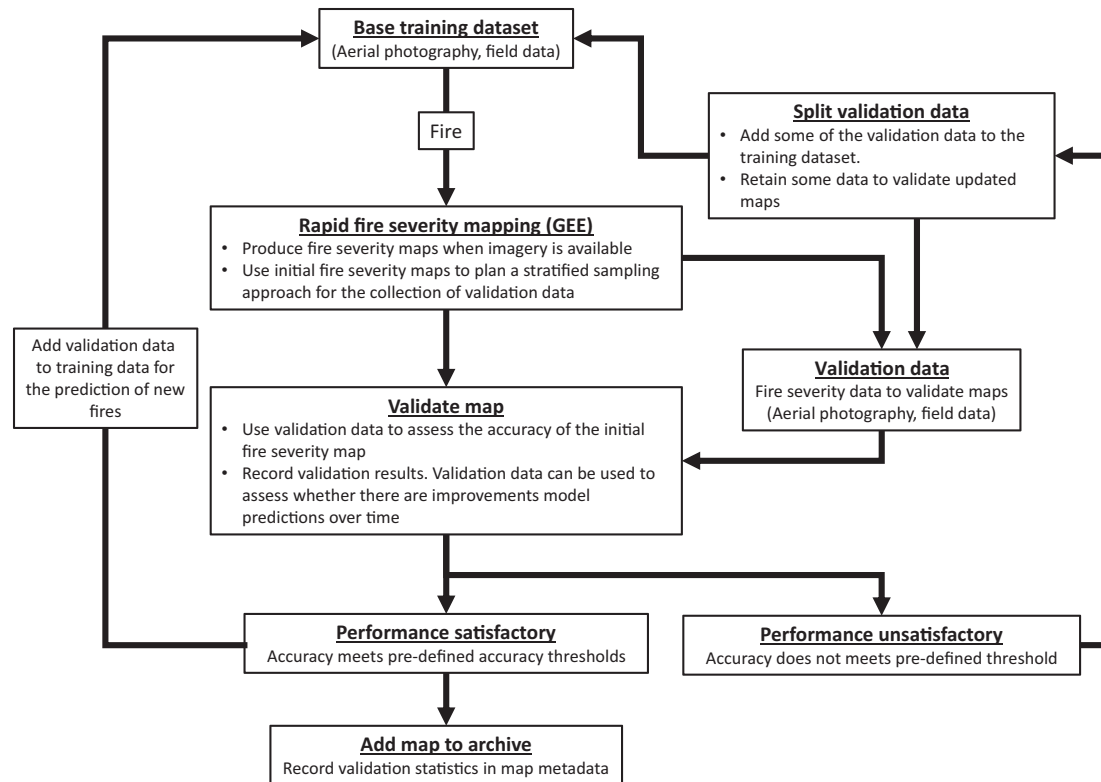


Fig. 8. A framework for the production of fire severity maps, assessment of map accuracy and continued refinement of classification. GEE is recommended for the rapid production of fire severity mapping (see also Parks et al., 2018), though we note that other platforms will also be suitable for this purpose.

scale at which spatial autocorrelation is present can be large (~500 m) for crown consuming fires (Taylor et al., 2014), which generally occur during extreme fire weather events (Bradstock et al., 2010). This distance decreases for lower fire severity classes (e.g. ~200 m when crown scorching fires are considered; Taylor et al., 2014), which generally occur under benign weather conditions and are influenced more so by topography and fuels (Bradstock et al., 2010). We used a minimum spatial separation of 85 m between data points, which is similar to other studies (e.g. 100 m; Meddens et al., 2016), and is likely sufficient spacing for low fire severity classes. Inflated predictive accuracy due to spatial autocorrelation is therefore only likely to be an issue for the CC severity class. We found that the RF model had high classification accuracy of CC (> 90%; Fig. 3) when applied to fires that had been held out of the training dataset, indicating that spatial autocorrelation was likely having little effect on the predictive performance of the RF model.

GEE provides an ideal platform for the rapid production of accurate fire severity maps using the RF classifier (Fig. 7), especially given the large volume of data, high level of data curation, and the computational efficiencies (Gorelick et al., 2017). The application of the RF classifier in the GEE platform rapidly produced and exported (i.e. < 15 mins processing time) severity maps for large wildfires (10000–100,000 ha). These maps had similar spatial patterns in fire severity classes as maps produced using time-consuming hand digitisation of aerial images. GEE and the embedded RF classifier are well suited to routine acquisition and continued refinement of fire severity mapping using Landsat and other moderate resolution sensors with regular return intervals (e.g. Sentinel). The capacity for the RF classifier to improve with the

addition of new training data will facilitate constant refinement of predictions over time. The development of a graphical user interface (GUI) to facilitate the extraction of spectral indices, validation of mapping and the production of fire severity maps will likely be necessary to increase the accessibility of this technique to non-technical users within fire management agencies.

We provide a framework for the integration of the RF mapping approach into fire management operations to routinely produce and improve fire severity maps (Fig. 8). The approach outlines a workflow for (i) the production of fire severity maps following fires, (ii) routine collection of fire severity data for initial map validation and necessary training, and (iii) continued assessment of the classifier accuracy to assess the need for the collection of additional training data (Fig. 8). The costs associated with the acquisition of training and validation data derived from field measures or aerial photography (McCarthy et al., 2017) may limit the number of fires that can be sampled for training and validation purposes. Thus, it may only be feasible for training and validation to be performed for a subset of wildfires. Collation of existing field derived fire severity datasets (e.g. CBI) (Kolden et al., 2015) could be used for the training and validation of a base RF model in some instances. Additionally, the rising popularity of drones and other unmanned aerial vehicles for the acquisition of aerial imagery (e.g. Meng et al., 2017) will likely reduce the costs of data collection in the future.

5. Conclusions

Accurate fire severity mapping is critical to improving understanding of changes to fire regimes over space and time, the

effectiveness of fire management actions, validation and refinement of fire behaviour models and better understanding of ecological effects of fire severity (e.g. Bennett et al., 2016; Eidenshink et al., 2007; Price and Bradstock, 2012). From a land management perspective, the rapid production of fire severity maps enables timely and cost-effective recovery, rehabilitation and restoration efforts. We demonstrate that the use of a RF classifier, with thoughtfully selected Δ FSI and pre-fire SI, leads to improved classification accuracy of fire severity data products, relative to widely adopted approaches using only a single Δ FSI, such as Δ NBR. In particular, the ability of the RF classifier to improve the discrimination of low and moderate fire severity classes (i.e. CU, PCS, CS) will be of considerable advantage for both ecological and fire behaviour studies. Improvements to classifier accuracy will lead to more useful classifications of recent fires, as well as increased accuracy of fire severity classifications for previous fire events. These improvements will aid post-fire management response and provide the opportunity for refining our understanding of wildfire behaviour and ecological responses to fire. The extensive archive of Landsat data in GEE also means this framework can be applied to more accurately and consistently map historic wildfire severity, thus improving fire history archives.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rse.2018.07.005>.

References

- Adler, D., 2005. *vioplot*: Violin plot. In: R package version 0.2, . <http://wsopuppenkiste.wiso.uni-goettingen.de/~dadler>.
- Barrett, K., McGuire, A.D., Hoy, E.E., Kasischke, E.S., 2011. Potential shifts in dominant forest cover in interior Alaska driven by variations in fire severity. *Ecol Appl* 21, 2380–2396.
- Belgiu, M., Drăguț, L., 2016. Random forest in remote sensing: a review of applications and future directions. *ISPRS J Photogramm Remote Sens* 114, 24–31.
- Bennett, L.T., Bruce, M.J., MacHunter, J., Kohout, M., Tanase, M.A., Aponte, C., 2016. Mortality and recruitment of fire-tolerant eucalypts as influenced by wildfire severity and recent prescribed fire. *For Ecol Manag* 380, 107–117.
- Boer, M.M., Macfarlane, C., Norris, J., Sadler, R.J., Wallace, J., Grierson, P.F., 2008. Mapping burned areas and burn severity patterns in SW Australian eucalypt forest using remotely-sensed changes in leaf area index. *Remote Sens Environ* 112, 4358–4369.
- Bowman, D.M.J.S., Balch, J.K., Artaxo, P., Bond, W.J., Carlson, J.M., Cochrane, M.A., D'Antonio, C.M., DeFries, R.S., Doyle, J.C., Harrison, S.P., Johnston, F.H., Keeley, J.E., Krawchuk, M.A., Kull, C.A., Marston, J.B., Moritz, M.A., Prentice, I.C., Roos, C.I., Scott, A.C., Swetnam, T.W., van der Werf, G.R., Pyne, S.J., 2009. Fire in the earth system. *Science* 324, 481–484.
- Bradstock, R.A., Hammill, K.A., Collins, L., Price, O., 2010. Effects of weather, fuel and terrain on fire severity in topographically diverse landscapes of south-eastern Australia. *Landscape Ecol* 25, 607–619.
- Breiman, L., 2001. Random forests. *Mach Learn* 45, 5–32.
- Brewer, C.K., Winne, J.C., Redmond, R.L., Opitz, D.W., Mangrich, M.V., 2005. Classifying and mapping wildfire severity. *Photogramm Eng Remote Sens* 71, 1311–1320.
- Caccamo, G., Chisholm, L.A., Bradstock, R.A., Puotinen, M.L., Phippen, B.G., 2012. Monitoring live fuel moisture content of heathland, shrubland and sclerophyll forest in south-eastern Australia using MODIS data. *Int J Wildland Fire* 21, 257–269.
- Chafer, C.J., Noonan, M., Macnaught, E., 2004. The post-fire measurement of fire severity and intensity in the Christmas 2001 Sydney wildfires. *Int J Wildland Fire* 13, 227–240.
- Champagne, C., McNairn, H., Daneshfar, B., Shang, J., 2014. A bootstrap method for assessing classification accuracy and confidence for agricultural land use mapping in Canada. *Int J Appl Earth Obs Geoinf* 29, 44–52.
- Chuvieco, E., Martín, M.P., Palacios, A., 2002. Assessment of different spectral indices in the red-near-infrared spectral domain for burned land discrimination. *Int J Remote Sens* 23, 5103–5110.
- Chuvieco, E., Riaño, D., Danson, F.M., Martín, P., 2006. Use of a radiative transfer model to simulate the postfire spectral response to burn severity. *J Geophys Res Biogeosci* 111 (n/a–n/a).
- Collins, B.M., Kelly, M., van Wagtenonk, J.W., Stephens, S.L., 2007. Spatial patterns of large natural fires in Sierra Nevada wilderness areas. *Landscape Ecol* 22, 545–557.
- Collins, L., Bradstock, R.A., Penman, T.D., 2014. Can precipitation influence landscape controls on wildfire severity? A case study within temperate eucalypt forests of south-eastern Australia. *Int J Wildland Fire* 23, 9–20.
- Coops, N., Delahaye, A., Pook, E., 1997. Estimation of eucalypt forest leaf area index on the south coast of New South Wales using Landsat MSS data. *Aust J Bot* 45, 757–769.
- Cutler, D.R., Edwards, T.C., Beard, K.H., Cutler, A., Hess, K.T., Gibson, J., Lawler, J.J., 2007. Random forests for classification in ecology. *Ecology* 88, 2783–2792.
- De Santis, A., Chuvieco, E., Vaughan, P.J., 2009. Short-term assessment of burn severity using the inversion of PROSPECT and GeoSail models. *Remote Sens Environ* 113, 126–136.
- De Santis, A., Asner, G.P., Vaughan, P.J., Knapp, D.E., 2010. Mapping burn severity and burning efficiency in California using simulation models and Landsat imagery. *Remote Sens Environ* 114, 1535–1545.
- DeSantis, A., Chuvieco, E., 2007. Burn severity estimation from remotely sensed data: performance of simulation versus empirical models. *Remote Sens Environ* 108, 422–435.
- Doerr, S.H., Shakesby, R.A., Blake, W.H., Chafer, C.J., Humphreys, G.S., Wallbrink, P.J., 2006. Effects of differing wildfire severities on soil wettability and implications for hydrological responses. *J Hydrol* 319, 295–311.
- Du, P., Samat, A., Waske, B., Liu, S., Li, Z., 2015. Random Forest and Rotation Forest for fully polarized SAR image classification using polarimetric and spatial features. *ISPRS J Photogramm Remote Sens* 105, 38–53.
- Ducroy, M., Duhoux, F., Huc, R., Rigolot, E., 1996. The ecophysiological and growth responses of Aleppo pine (*Pinus halepensis*) to controlled heating applied to the base of the trunk. *Can J For Res* 26, 1366–1374.
- Eidenshink, J., Schwind, B., Brewer, K., Zhu, Z., Quayle, B., Howard, S., 2007. A project for monitoring trends in burn severity. *Fire Ecol* 3, 3–21 (Special Issue).
- Epting, J., Verbyla, D., Sorbel, B., 2005. Evaluation of remotely sensed indices for assessing burn severity in interior Alaska using Landsat TM and ETM+. *Remote Sens Environ* 96, 328–339.
- Gao, B.-C., 1996. NDWI—a normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sens Environ* 58, 257–266.
- Ghimire, B., Rogan, J., Galiano, V.R., Panday, P., Neeti, N., 2012. An evaluation of bagging, boosting, and random forests for land-cover classification in Cape Cod, Massachusetts, USA. *GISci. Remote Sens.* 49, 623–643.
- Gitelson, A.A., Kaufman, Y.J., Stark, R., Rundquist, D., 2002. Novel algorithms for remote estimation of vegetation fraction. *Remote Sens Environ* 80, 76–87.
- Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sens Environ* 202, 18–27.
- Graves, S., Asner, G., Martin, R., Anderson, C., Colgan, M., Kalantari, L., Bohlman, S., 2016. Tree species abundance predictions in a tropical agricultural landscape with a supervised classification model and imbalanced data. *Remote Sens* 8, 161.
- Hammill, K.A., Bradstock, R.A., 2006. Remote sensing of fire severity in the Blue Mountains: influence of vegetation type and inferring fire intensity. *Int J Wildland Fire* 15, 213–226.
- Harris, S., Veraverbeke, S., Hook, S., 2011. Evaluating spectral indices for assessing fire severity in chaparral ecosystems (Southern California) using MODIS/ASTER (MASTER) airborne simulator data. *Remote Sens* 3, 2403.
- Hawbaker, T.J., Vanderhoof, M.K., Beal, Y.-J., Takacs, J.D., Schmidt, G.L., Falgout, J.T., Williams, B., Fairaux, N.M., Caldwell, M.K., Picotte, J.J., Howard, S.M., Stitt, S., Dwyer, J.L., 2017. Mapping burned areas using dense time-series of Landsat data. *Remote Sens Environ* 198, 504–522.
- Heydari, S.S., Mountrakis, G., 2018. Effect of classifier selection, reference sample size, reference class distribution and scene heterogeneity in per-pixel classification accuracy using 26 Landsat sites. *Remote Sens Environ* 204, 648–658.
- Holden, Z.A., Smith, A.M.S., Morgan, P., Rollins, M.G., Gessler, P.E., 2005. Evaluation of novel thermally enhanced spectral indices for mapping fire perimeters and comparisons with fire atlas data. *Int J Remote Sens* 26, 4801–4808.
- Holden, Z.A., Morgan, P., Evans, J.S., 2009. A predictive model of burn severity based on 20-year satellite-inferred burn severity data in a large southwestern US wilderness area. *For Ecol Manag* 258, 2399–2406.
- Hultquist, C., Chen, G., Zhao, K., 2014. A comparison of Gaussian process regression, random forests and support vector regression for burn severity assessment in diseased forests. *Remote Sens. Lett.* 5, 723–732.
- Keeley, J.E., 2009. Fire intensity, fire severity and burn severity: a brief review and suggested usage. *Int J Wildland Fire* 18, 116–126.
- Key, C.H., Benson, N.C., 2006. Landscape assessment (LA). In: Lutes, D.C., Keane, R.E., Caratti, J.F., Key, C.H., Benson, N.C., Sutherland, S., Gangi, L.J. (Eds.), FIREMON: Fire Effects Monitoring and Inventory System. U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station, Fort Collins, CO.
- Knox, K.J.E., Clarke, P.J., 2012. Fire severity, feedback effects and resilience to alternative community states in forest assemblages. *For Ecol Manag* 265, 47–54.
- Kolden, C.A., Smith, A.M.S., Abatzoglou, J.T., 2015. Limitations and utilisation of Monitoring Trends in Burn Severity products for assessing wildfire severity in the USA. *Int J Wildland Fire* 24, 1023–1028.
- Krawchuk, M.A., Haire, S.L., Coop, J., Parisien, M.-A., Whitman, E., Chong, G., Miller, C., 2016. Topographic and fire weather controls of fire refugia in forested ecosystems of northwestern North America. *Ecosphere* 7 (e01632-n/a).
- Kuhn, M., 2008. Building predictive models in R using the caret package. *J Stat Softw* 28,

- 1–26.
- Lentile, L.B., Holden, Z.A., Smith, A.M.S., Falkowski, M.J., Hudak, A.T., Morgan, P., Lewis, S.A., Gessler, P.E., Benson, N.C., 2006. Remote sensing techniques to assess active fire characteristics and post-fire effects. *Int J Wildland Fire* 15, 319–345.
- Leonard, S.W.J., Bennett, A.F., Clarke, M.F., 2014. Determinants of the occurrence of unburnt forest patches: potential biotic refuges within a large, intense wildfire in south-eastern Australia. *For Ecol Manag* 314, 85–93.
- Liaw, A., Wiener, M., 2002. Classification and regression by randomForest. *R News* 2, 18–22.
- McCarthy, G., Moon, K., Smith, L., 2017. Mapping fire severity and fire extent in forest in Victoria for ecological and fuel outcomes. *Ecol Manag Restor* 18, 54–65.
- Meddens, A.J.H., Kolden, C.A., Lutz, J.A., 2016. Detecting unburned areas within wildfire perimeters using Landsat and ancillary data across the northwestern United States. *Remote Sens Environ* 186, 275–285.
- Mellor, A., Boukir, S., 2017. Exploring diversity in ensemble classification: applications in large area land cover mapping. *ISPRS J Photogramm Remote Sens* 129, 151–161.
- Mellor, A., Boukir, S., Haywood, A., Jones, S., 2015. Exploring issues of training data imbalance and mislabelling on random forest performance for large area land cover classification using the ensemble margin. *ISPRS J Photogramm Remote Sens* 105, 155–168.
- Meng, R., Wu, J., Schwager, K.L., Zhao, F., Dennison, P.E., Cook, B.D., Brewster, K., Green, T.M., Serbin, S.P., 2017. Using high spatial resolution satellite imagery to map forest burn severity across spatial scales in a Pine Barrens ecosystem. *Remote Sens Environ* 191, 95–109.
- Millard, K., Richardson, M., 2015. On the importance of training data sample selection in random forest image classification: a case study in peatland ecosystem mapping. *Remote Sens* 7, 8489.
- Miller, J.D., Knapp, E.E., Key, C.H., Skinner, C.N., Isbell, C.J., Creasy, R.M., Sherlock, J.W., 2009. Calibration and validation of the relative differenced Normalized Burn Ratio (RdNBR) to three measures of fire severity in the Sierra Nevada and Klamath Mountains, California, USA. *Remote Sens Environ* 113, 645–656.
- Nolan, R.H., Lane, P.N.J., Benyon, R.G., Bradstock, R.A., Mitchell, P.J., 2015. Trends in evapotranspiration and streamflow following wildfire in resprouting eucalypt forests. *J Hydrol* 524, 614–624.
- Nolan, R.H., Boer, M.M., Resco de Dios, V., Caccamo, G., Bradstock, R.A., 2016. Large-scale, dynamic transformations in fuel moisture drive wildfire activity across south-eastern Australia. *Geophys Res Lett* 43, 4229–4238.
- Parker, B.M., Lewis, T., Srivastava, S.K., 2015. Estimation and evaluation of multi-decadal fire severity patterns using Landsat sensors. *Remote Sens Environ* 170, 340–349.
- Parks, S., Holsinger, L., Voss, M., Loehman, R., Robinson, N., 2018. Mean composite fire severity metrics computed with Google Earth Engine offer improved accuracy and expanded mapping potential. *Remote Sens* 10, 879.
- Price, O.F., Bradstock, R.A., 2012. The efficacy of fuel treatment in mitigating property loss during wildfires: insights from analysis of the severity of the catastrophic fires in 2009 in Victoria, Australia. *J Environ Manag* 113, 146–157.
- R Development Core Team, 2016. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
- Ramo, R., Chuvieco, E., 2017. Developing a random forest algorithm for MODIS global burned area classification. *Remote Sens* 9, 1193.
- Robinson, N.M., Leonard, S.W.J., Ritchie, E.G., Bassett, M., Chia, E.K., Buckingham, S., Gibb, H., Bennett, A.F., Clarke, M.F., 2013. REVIEW: refuges for fauna in fire-prone landscapes: their ecological function and importance. *J Appl Ecol* 50, 1321–1329.
- Rodriguez-Galiano, V.F., Ghimire, B., Rogan, J., Chica-Olmo, M., Rigol-Sanchez, J.P., 2012. An assessment of the effectiveness of a random forest classifier for land-cover classification. *ISPRS J Photogramm Remote Sens* 67, 93–104.
- Rogan, J., Franklin, J., Stow, D., Miller, J., Woodcock, C., Roberts, D., 2008. Mapping land-cover modifications over large areas: a comparison of machine learning algorithms. *Remote Sens Environ* 112, 2272–2283.
- Roy, D.P., Boschetti, L., Trigg, S.N., 2006. Remote sensing of fire severity: assessing the performance of the normalized burn ratio. *IEEE Geosci Remote Sens Lett* 3, 112–116.
- Roy, D.P., Kovalsky, V., Zhang, H.K., Vermote, E.F., Yan, L., Kumar, S.S., Egorov, A., 2016. Characterization of Landsat-7 to Landsat-8 reflective wavelength and normalized difference vegetation index continuity. *Remote Sens Environ* 185, 57–70.
- Schmidt, G., Jenkerson, C., Masek, J., Vermote, E., Gao, F., 2013. Landsat ecosystem disturbance adaptive processing system (LEDAPS) algorithm description. In: Open-File Report. U. S. Geological Survey, Reston, VA, pp. 27.
- Smith, A.M.S., Sparks, A.M., Kolden, C.A., Abatzoglou, J.T., Talhelm, A.F., Johnson, D.M., Boschetti, L., Lutz, J.A., Apostol, K.G., Yedinak, K.M., Tinkham, W.T., Kremens, R.J., 2016. Towards a new paradigm in fire severity research using dose–response experiments. *Int J Wildland Fire* 25, 158–166.
- Smucker, K.M., Hutto, R.L., Steele, B.M., 2005. Changes in bird abundance after wildfire: importance of fire severity and time since fire. *Ecol Appl* 15, 1535–1549.
- Taylor, C., McCarthy, M.A., Lindenmayer, D.B., 2014. Nonlinear effects of stand age on fire severity. *Conserv Lett* 7, 355–370.
- Thompson, J.R., Spies, T.A., Ganio, L.M., 2007. Reburn severity in managed and unmanaged vegetation in a large wildfire. *PNAS* 104, 10743–10748.
- Trigg, S., Flasse, S., 2001. An evaluation of different bi-spectral spaces for discriminating burned shrub-savannah. *Int J Remote Sens* 22, 2641–2647.
- Tucker, C.J., 1979. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sens Environ* 8, 127–150.
- Veraverbeke, S., Verstraeten, W.W., Lhermitte, S., Goossens, R., 2010. Evaluating Landsat Thematic Mapper spectral indices for estimating burn severity of the 2007 Peloponnese wildfires in Greece. *Int J Wildland Fire* 19, 558–569.
- Veraverbeke, S., Lhermitte, S., Verstraeten, W.W., Goossens, R., 2011. Evaluation of pre/post-fire differenced spectral indices for assessing burn severity in a Mediterranean environment with Landsat Thematic Mapper. *Int J Remote Sens* 32, 3521–3537.
- White, J.D., Ryan, K.C., Key, C.C., Running, S.W., 1996. Remote sensing of forest fire severity and vegetation recovery. *Int J Wildland Fire* 6, 125–136.
- Yebra, M., Quan, X., Riaño, D., Rozas Larraondo, P., van Dijk, A.I.J.M., Cary, G.J., 2018. A fuel moisture content and flammability monitoring methodology for continental Australia based on optical remote sensing. *Remote Sens Environ* 212, 260–272.